CSEC Mathematics

January 2016 – Paper 2

Solutions

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SECTION I

Answer ALL questions in this section.

All working must be clearly shown.

1. (a) Using a calculator, or otherwise, calculate the EXACT value of

$$\left(3.6 + \sqrt{51.84}\right) \div 3.75$$ [2]

$$\left(3.6 + \sqrt{51.84}\right) \div 3.75$$

$$= \left(3.6 + \frac{36}{5}\right) \div 3.75$$

$$= \left(\frac{18}{5} + \frac{36}{5}\right) \div \frac{15}{4}$$

$$= \left(\frac{54}{5}\right) \div \frac{15}{4}$$

$$= \frac{54}{5} \times \frac{4}{15}$$

$$= \frac{72}{25}$$ [Exact Value]

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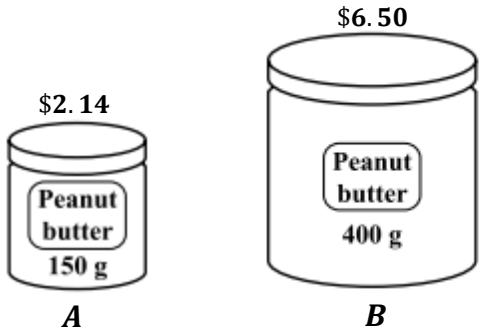
(b) The diagram below, **not drawn to scale**, shows two jars of peanut butter of the same brand.



Which of the jars shown above is the BETTER buy?

Show ALL working to support your answer. [3]

Jar A

$150g = \$2.14$

$1g = \frac{2.14}{150}$

$1g = 0.0143$ (correct to 3 significant figures)

Jar B

$400g = \$6.50$

$1g = \frac{6.50}{400}$

$1g = 0.0163$ (correct to 3 significant figures)

Comparing the price per gram for both jars, Jar A $<$ Jar B. Therefore, Jar A is the better buy.

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(c) Thomas invested \$1498 at 6% simple interest per annum.

Calculate:

(i) The interest, in dollars, earned after six months [2]

Time = 6 months

$$= \frac{6}{12} \text{ year}$$

$$= \frac{1}{2} \text{ or } 0.5 \text{ year}$$

Substituting Principal = \$1498, Rate = 6% and Time = 0.5 year into:

$$\text{Simple Interest} = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$$

$$= \frac{1498 \times 6 \times 0.5}{100}$$

$$= \$44.94$$

∴ After six months, the interest earned is \$44.94.

(ii) The TOTAL amount of money in his account after 3 years [2]

Substituting Principal = \$1498, Rate = 6% and Time = 3 years into:

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$$\text{Simple Interest} = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$$

$$= \frac{1498 \times 6 \times 3}{100}$$

$$= \$269.64$$

$$\text{Total amount after 3 years} = \text{Principal} + \text{Interest Earned}$$

$$= \$1498 + \$269.64$$

$$= \$1767.64$$

(iii) How long it will be before his investment earns \$449.40 [2]

Substituting Simple Interest = \$449.40, Principal = \$1498 and

Rate = 6% into:

$$\text{Time} = \frac{\text{Simple Interest} \times 100}{\text{Principal} \times \text{Rate}}$$

$$= \frac{449.40 \times 100}{1498 \times 6}$$

$$= 5 \text{ years}$$

∴ In 5 years, his investment will earn \$449.40.

Total: 11 marks

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2. (a) (i) Solve for x, where x is a real number.

$$8 - x \leq 5x + 2$$ [3]

$$8 - x \leq 5x + 2$$

$$8 - 2 \leq 5x + x$$

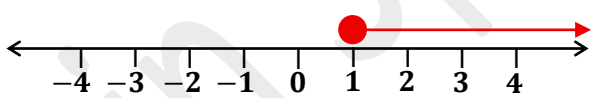
$$6 \leq 6x$$

$$6x \geq 6$$

$$x \geq \frac{6}{6}$$

$$x \geq 1$$

(ii) Show your solution to (a)(i) on the number line below. [1]



(b) Expand and simplify

$$2x(x + 5) - 3(x - 4)$$ [2]

$$2x(x + 5) - 3(x - 4)$$

$$= 2x^2 + 10x - 3x + 12$$

$$= 2x^2 + 7x + 12$$

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(c) Simplify

$$\frac{3x^2 \times 4x^3}{2x}$$ [2]

$$\frac{3x^2 \times 4x^3}{2x}$$

$$= \frac{12x^{2+3}}{2x}$$

$$= \frac{12x^5}{2x}$$

$$= \frac{12}{2}x^{5-1}$$

$$= 6x^4$$

(d) Write as a single fraction, in its lowest terms,

$$\frac{x+1}{2} + \frac{5-x}{5}$$ [2]

$$\frac{(x+1)}{2} + \frac{(5-x)}{5}$$

$$= \frac{5(x+1) + 2(5-x)}{10}$$

$$= \frac{5x + 5 + 10 - 2x}{10}$$

$$= \frac{5x - 2x + 5 + 10}{10}$$

$$= \frac{3x + 15}{10}$$

$$= \frac{3(x+5)}{10}$$

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(e) Factorize completely

$4x^2 - 4$ [2]

$4x^2 - 4$

$= (2x)^2 - (2)^2$ [Difference of two squares]

$= (2x - 2)(2x + 2)$

Total: 12 marks

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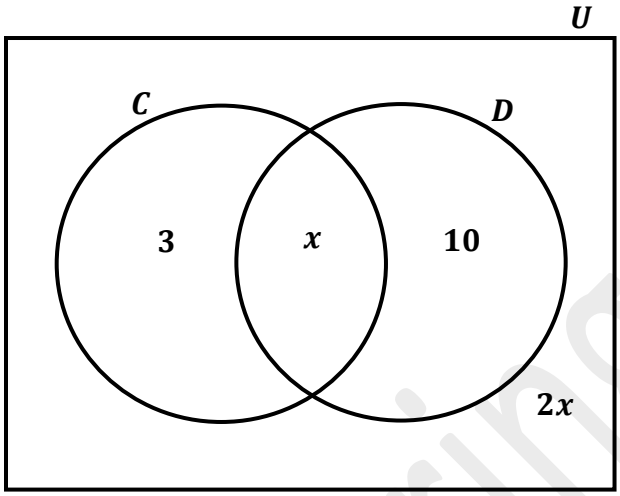
3. (a) The Venn diagram shows the number of students in Form 5A who have visited Canada (C) or Dominica (D).



(i) How many students have visited Dominica ONLY? [1]

Number of students who visited Dominica only $= 10$

(ii) Write an **expression**, in terms of x, to represent the TOTAL number of students who have visited Canada. [1]

Total number of students who have visited Canada $= 3 + x$

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(iii) Given that there are 25 students in Form 5A, calculate the value of x. [2]

Total number of students in Form 5A = 250

$$3 + x + 10 + 2x = 25$$

$$13 + 3x = 25$$

$$3x = 25 - 13$$

$$3x = 12$$

$$x = \frac{12}{3}$$

$$x = 4$$

(iv) Hence, write down the number of students in each of the following subsets: [3]

- $C \cup D$

  $= 3 + x + 10$

  $= 3 + (4) + 10$

  $= 17$

- $C \cap D$

  $= (x)$

  $= 4$

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- $(C \cup D)'$

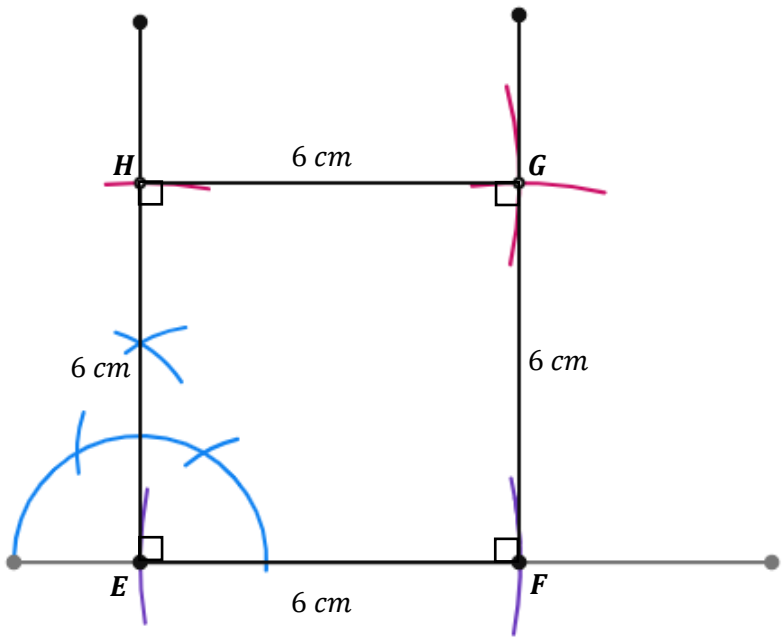
  $= 2x$

  $= 2(4)$

  $= 8$

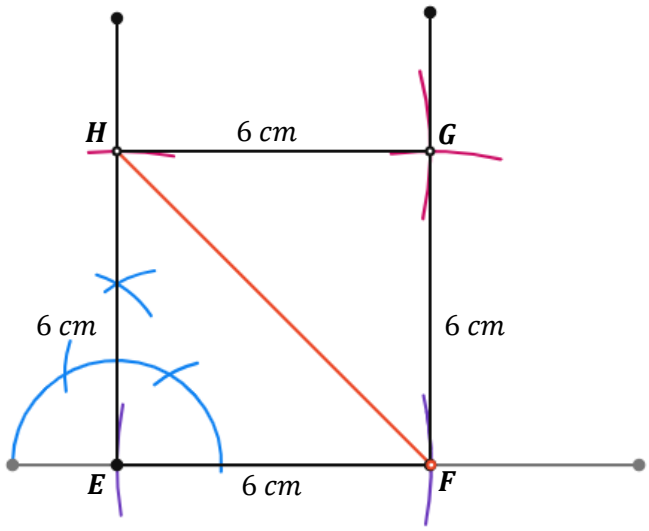
(b) (i) Using **a ruler, a pencil and a pair of compasses**, construct accurately, the **square** $EFGH$ where $EF = 6\ cm$.

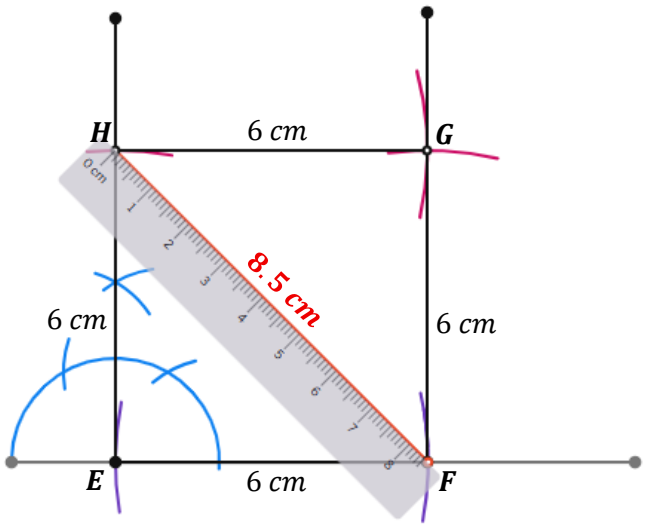
(Show ALL construction lines and curves.) [4]



(ii) Measure, and state in centimetres, the length of the diagonal FH.

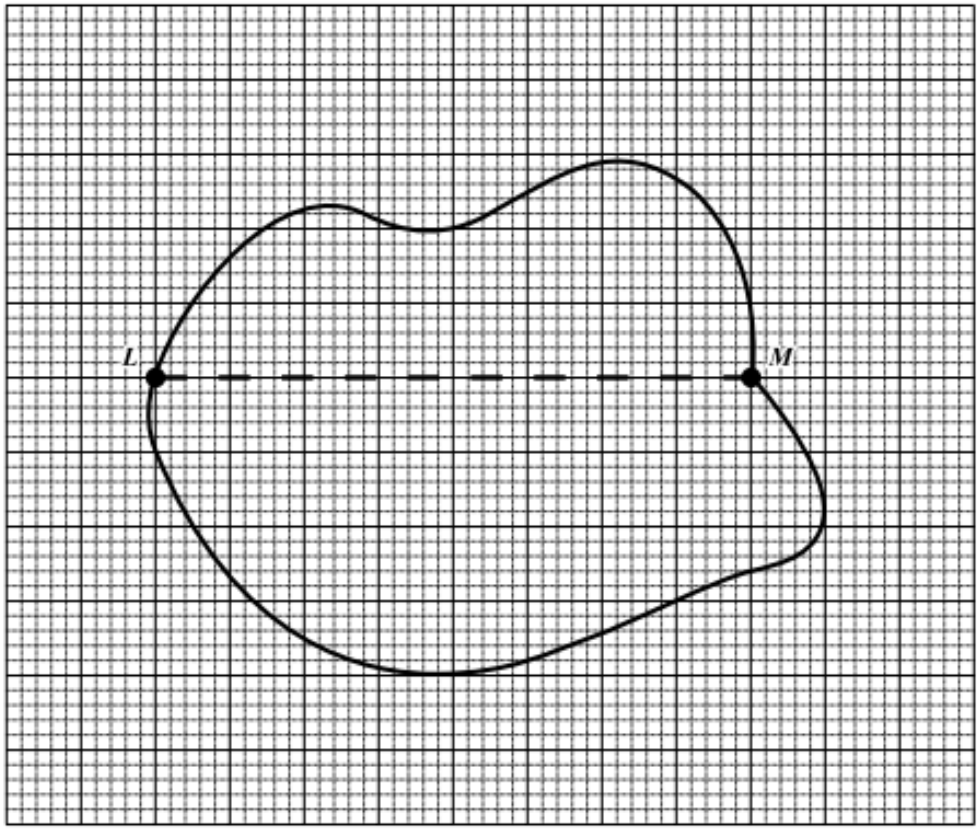
$FH =$ 8.5 cm [1]





Total: 12 marks

4. (a) The diagram below shows a map of an island drawn on a grid of 1 cm squares.



(i) State, in cm, the length of LM as shown in the diagram.

$LM =$ ……8…… cm [1]

The length of LM on the diagram $= 8\ cm$.

(ii) Estimate, by counting squares, the area of the map shown in the diagram. [1]

Note:

An area that occupies <u>more than half</u> of the square is considered as 1 square. An area that occupies <u>less than half</u> of the square is not counted.

$\therefore$ The total number of squares in the diagram $= 45$.

Since each square has an area of $1\ cm^2$,

The total area of the map $= 45\ cm^2$

(iii) On the island, the actual distance LM is $20\ km$. Complete the following statement:

On the map, $1\ cm$ represents ……**2.5**…… km. [1]

The actual distance of $LM = 20\ km$

On the map, LM has a distance of $8\ cm$.

$\therefore 8\ cm = 20\ km$

$$1\ cm = \frac{20}{8}\ km$$

$$= 2.5\ km$$

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(iv) Write the scale of the map in the form $1 : x$. [1]

$1\ cm = 2.5\ km$

$= 2.5\ \times (100\ 000)\ cm$

$= 250\ 000\ cm$

$\therefore$ The scale of the map is $1: 250\ 000$ where $x = 250\ 000$.

(v) What distance on the island will be 3 cm on the map? [1]

$3\ cm = 3 \times 2.5\ km$

$= 7.5\ km$

(vi) What area on the island will be represented by $3\ cm^2$ on the map? [2]

$1\ cm = 2.5\ km$

$\therefore 1\ cm^2 = (2.5)^2 km^2$

$= 6.25\ km^2$

$3\ cm^2 = 3 \times 6.25\ km^2$

$= 18.75\ km^2$

$\therefore 18.75\ km^2$ will be represented by $3\ cm^2$ on the map.

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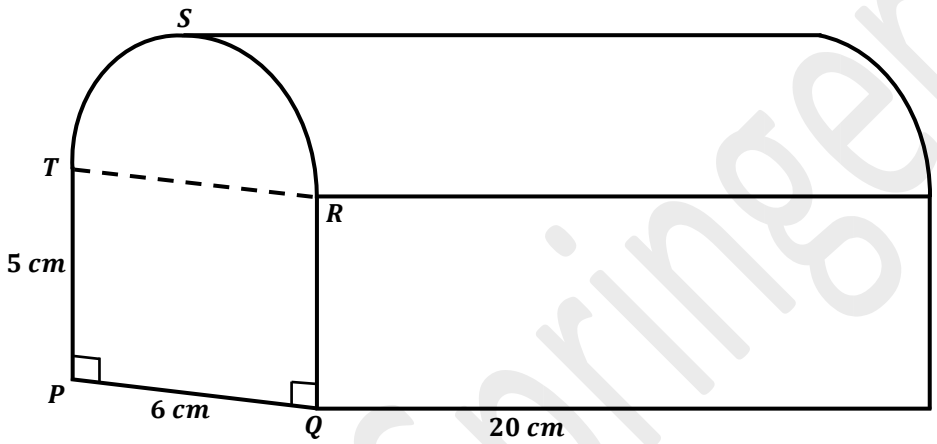
(b) The diagram below, **not drawn to scale**, shows a prism with cross section $PQRST$ and length 20 cm. $PQRST$ is made up of a rectangle $PQRT$ and a semicircle RST such that $PQ = 6$ cm and $QR = 5$ cm.

Use $\pi = 3.14$



(i) Calculate the area of the cross section $PQRST$. [2]

$$Area\ of\ rectangle\ PQRT = length \times width$$

$$= 6\ cm \times 5\ cm$$

$$= 30\ cm^2$$

$$Area\ of\ semi\text{-}circle\ RST = \frac{1}{2}\pi r^2$$

$$= \frac{1}{2}\ (3.14)(3)^2$$

$$= 14.13\ cm^2$$

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$$Area\ of\ PQRST = Area\ of\ rectangle\ PQRT + Area\ of\ semi\text{-}circle\ RST$$

$$= 30\ cm^2 + 14.13\ cm^2$$

$$= 44.13\ cm^2$$

(ii) An engineer needs a similar prism whose volume is NOT more than 900 cm^3.

Estimate, in cm, the length of the longest prism he can use. [2]

Let the length of the prism $= l$

<u>Recall:</u>

$$Volume\ of\ prism = Cross\text{-}sectional\ area \times length$$

$$Cross\text{-}sectional\ area \times length = Volume\ of\ prism$$

$$length = \frac{Volume\ of\ prism}{Cross - sectional\ area}$$

$$l \leq \frac{900\ cm^3}{44.13\ cm^2}$$

$$l \leq 20.4\ cm \qquad \text{[to 3 significant figures]}$$

∴ The length of the longest prism he can use is 20.4 cm.

Total: 11 marks

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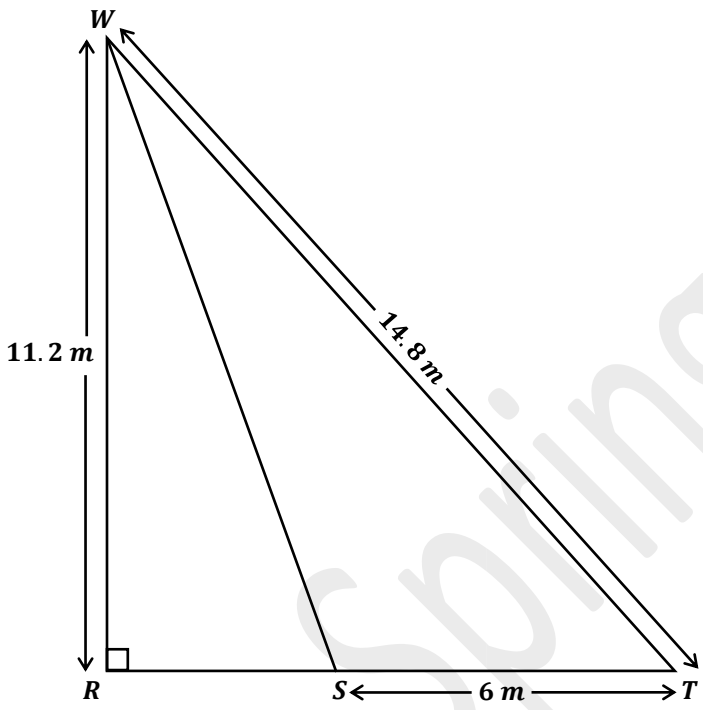
5. (a) In the diagram below, **not drawn to scale**, $ST = 6\ m$, $WR = 11.2\ m$, $WT = 14.8\ m$ and angle $WRS = 90°$.



Calculate, giving your answer to 1 decimal place

(i) the length RS [2]

Using Pythagoras' Theorem,

$(WR)^2 + (RT)^2 = (WT)^2$

$(11.2)^2 + (RT)^2 = (14.8)^2.$

$(RT)^2 = (14.8)^2 - (11.2)^2$

$(RT)^2 = 93.6$

$RT = \sqrt{93.6}$

$RT = 9.7\ m$

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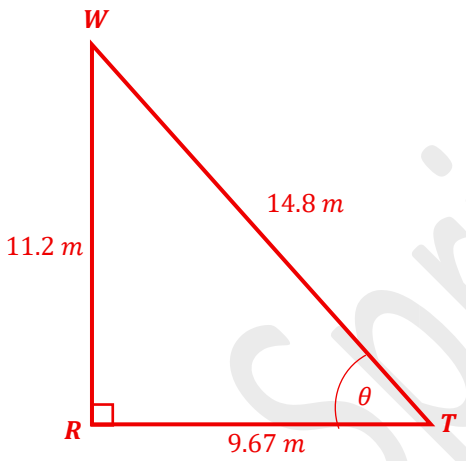
$RS = RT - ST$

$= 9.7 - 6$

$= 3.7\ m$ [correct to 1 decimal place]

(ii) the measure of angle RTW [2]



Let $R\hat{T}W = \theta$

$$\sin\theta = \frac{11.2}{14.8}$$

$$\theta = \sin^{-1}\left(\frac{11.2}{14.8}\right)$$

$\theta = 49.2°$ [correct to 1 decimal place]

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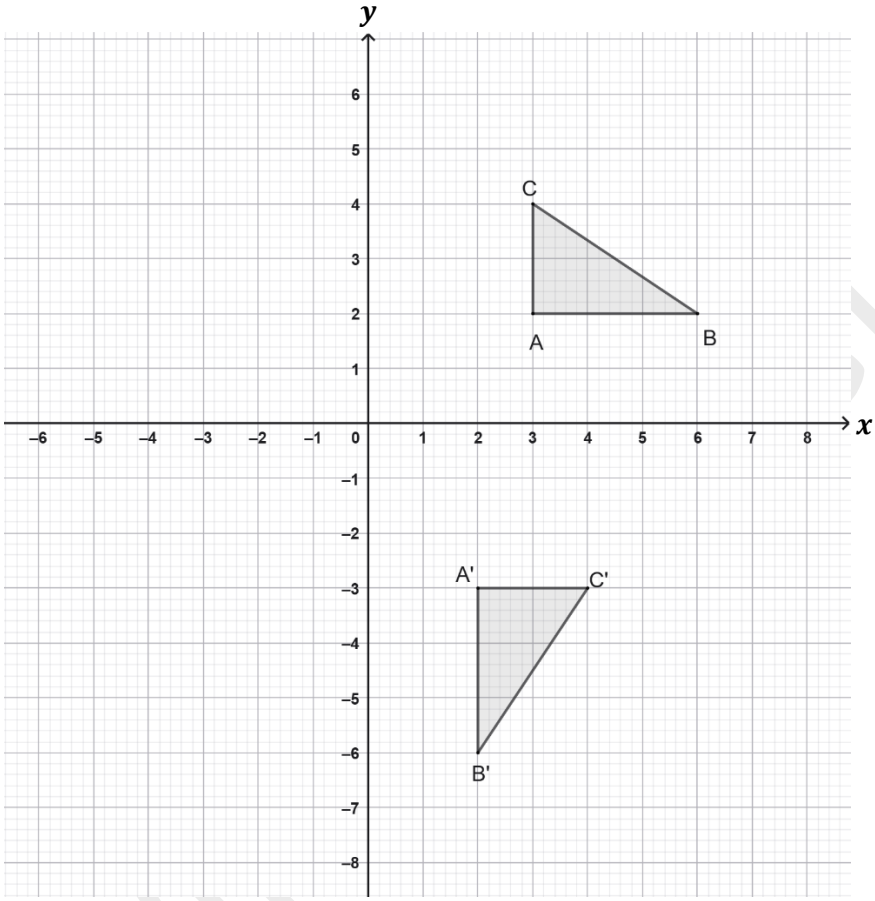
(b) The graph below shows a triangle ABC and its image $A'B'C'$ after undergoing a single transformation.



(i) Write down the coordinates of the vertices of ΔABC. [1]

$A = (3, 2)$

$B = (6, 2)$

$C = (3, 4)$

(ii) Write down the coordinates of the vertices of $\Delta A'B'C'$. [1]

$A' = (2, -3)$

$B' = (2, -6)$

$C' = (4, -3)$

(iii) Describe FULLY the transformation that maps triangle ABC onto triangle $A'B'C'$. [2]

The transformation that maps triangle ABC onto triangle $A'B'C'$ is a 90° clockwise rotation about O.

(iv) On the graph on page 16, draw the line $x = 1$ AND the triangle $A"B"C"$, the image of triangle ABC after a reflection in the line $x = 1$. [3]

$A'' = (-1, 2)$

$B'' = (-4, 2)$

$C'' = (-1, 4)$

The line $x = 1$ and the triangle $A"B"C"$ is shown on the graph below.

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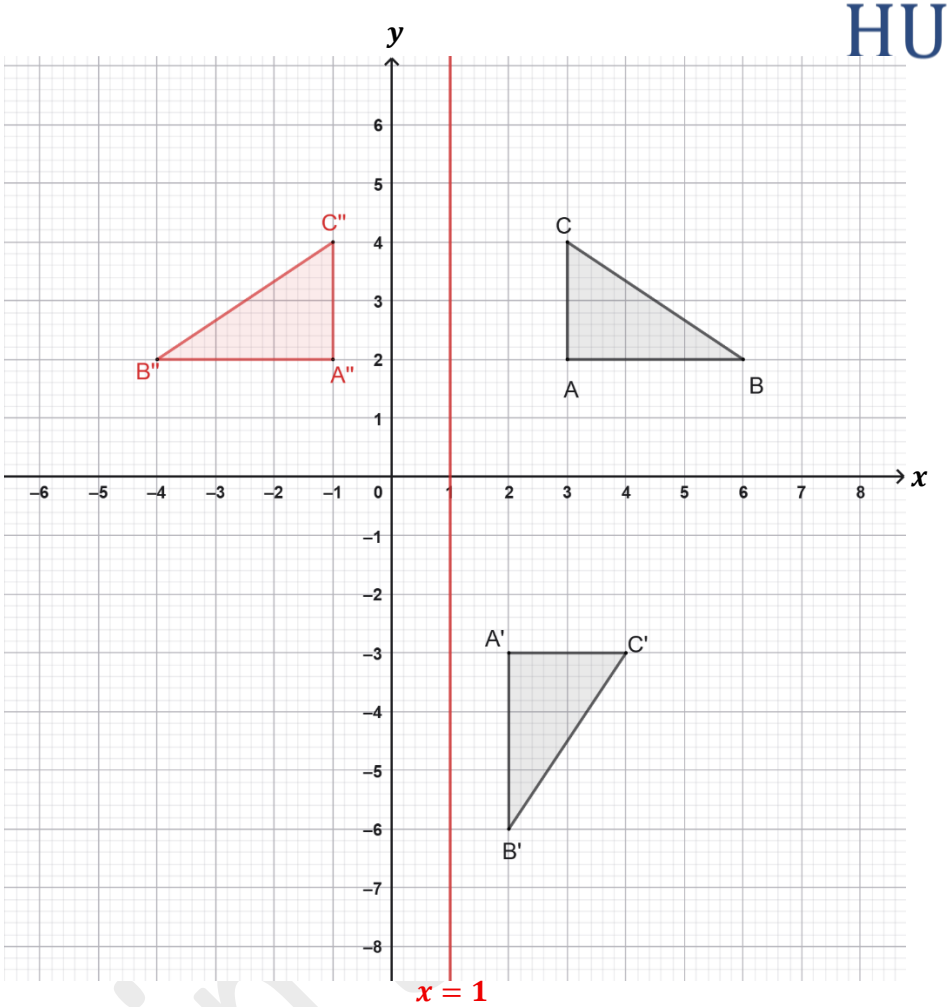


(v) State ONE geometrical relationship among $\Delta ABC, \Delta A'B'C'$ and $\Delta A"B"C"$.

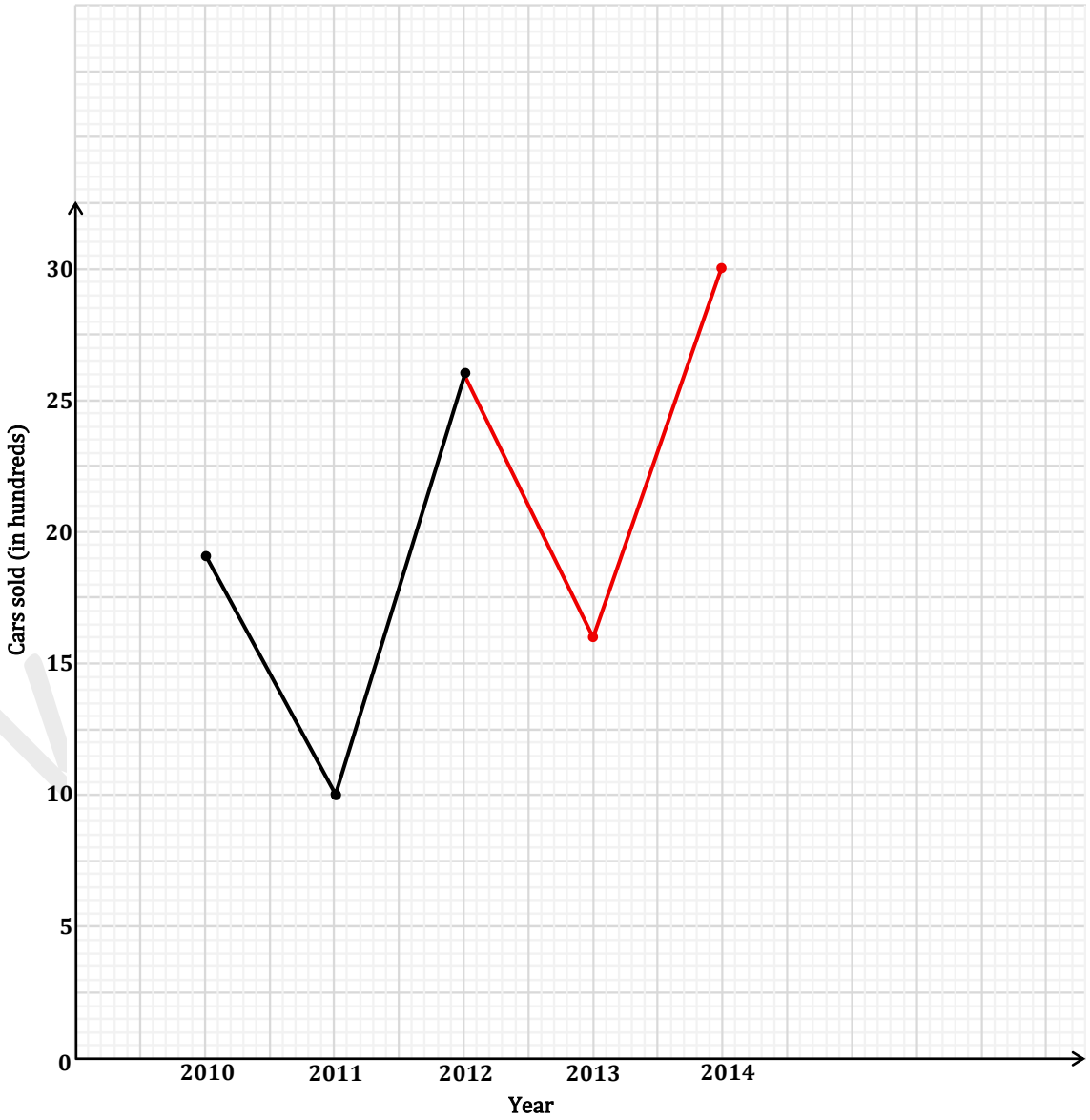
[1]

$\Delta ABC, \Delta A'B'C'$ and $\Delta A"B"C"$ are congruent.

Total: 12 marks

6. (a) The table below gives the number of cars sold in a country, in hundreds, from 2010 to 2014.

Year	2010	2011	2012	2013	2014
Cars sold (in hundreds)	19	10	26	16	30



(i) Complete the line graph on page 18 to represent the given information.

[1]

The line graph was completed above.

(ii) Between which two consecutive years was there the GREATEST increase in cars sold? [1]

The greatest increase in cars was sold between 2011 and 2012.

(iii) What was the TOTAL number of cars sold in five year period 2010 to 2014? [2]

Year	Cars Sold
2010	1900
2011	1000
2012	2600
2013	1600
2014	3000
Total =	**10100**

∴ The total number of cars sold from 2010 to 2014 was 10,100 cars.

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(iv) The mean number of cars sold from **2010** to **2015** was 22.5 hundred.

How many cars were sold in 2015? [2]

Mean number of cars sold from 2010 to 2015 $= 22.5 \times 100$

$= 2250\ cars$

NOTE: This is the <u>average</u> number of cars sold (per year) during the period 2010 to 2015.

$\therefore$ Total number of cars sold from 2010 to 2015 $= 2250\ \times 6$

$= 13,500\ cars$

From part (iii), we found the total number of cars sold from 2010 to 2014 was 10,100 cars.

$\therefore$ Number of cars sold in 2015

$=$ Num of cars sold (2010 to 2015) $-$ Num of cars sold (2010 to 2014)

$= 13{,}500 - 10{,}100$

$= 3400\ cars$

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(b) (i) A line JK has equation $2y = 5x + 6$. Determine the gradient of JK. [2]

Gradient of the line JK is ……$\frac{5}{2}$…… .

$2y = 5x + 6$

$y = \frac{5x + 6}{2}$

$y = \frac{5}{2}x + \frac{6}{2}$

$y = \frac{5}{2}x + 3$

Which is of the form $y = mx + c$

Where the gradient, $m = \frac{5}{2}$

Another line GH, is perpendicular to JK and passes through the point $(5, -1)$.

(ii) State the gradient of the line GH. [1]

Gradient of the line GH is ……$-\frac{2}{5}$…… .

Since the line GH is perpendicular to JK, then the gradient of GH will be the negative reciprocal of the gradient of JK.

$\therefore$ The gradient of the line GH is $-\frac{2}{5}$

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(iii) Determine the equation of line GH. [2]

Equation of the line GH is …….. .

Point that passes through GH $= (5, -1)$

Gradient, m of GH $= -\frac{2}{5}$

Equation of GH:

$$y - y_1 = m(x - x_1)$$

$$y - (-1) = -\frac{2}{5}(x - 5)$$

$$y + 1 = -\frac{2}{5}x + 2$$

$$y = -\frac{2}{5}x + 2 - 1$$

$$y = -\frac{2}{5}x + 1$$

$\therefore$ The equation of GH is $y = -\frac{2}{5}x + 1$

Total: 11 marks

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7. The table below shows how the minutes taken by all students to complete a science experiment were recorded and grouped.

Time (minutes)	Number of Students who Completed (Frequency)	Cumulative Frequency
1 – 5	1	1
6 – 10	2	3
11 – 15	5	8
16 – 20	7	15
21 – 25	10	25
26 – 30	15	40
31 – 35	8	48
36 – 40	2	50

(a) Complete the cumulative frequency column in the table. [2]

Shown in the table above.

(b) On the grid on page 23, using a scale of **2 cm to represent 5 minutes on the x-axis** and **2 cm to represent 5 students on the y-axis**, draw a cumulative frequency curve to represent the information in the table. [5]

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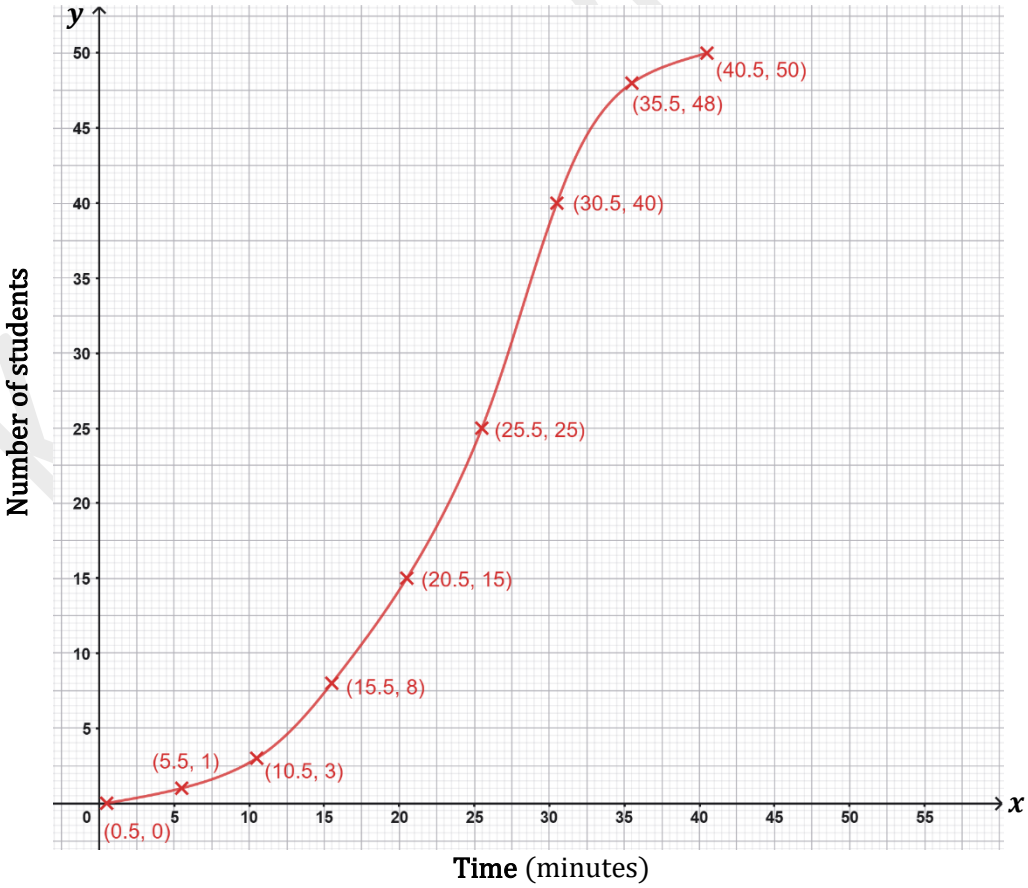
Time, t (minutes)	L.C.B $\leq t <$ U.C.B.	Cumulative Frequency	Points to be plotted
$1-5$	$0.5 \leq t < 5.5$	1	$(5.5, 1)$
$6-10$	$5.5 \leq t < 10.5$	3	$(10.5, 3)$
$11-15$	$10.5 \leq t < 15.5$	8	$(15.5, 8)$
$16-20$	$15.5 \leq t < 20.5$	15	$(20.5, 15)$
$21-25$	$20.5 \leq t < 25.5$	25	$(25.5, 25)$
$26-30$	$25.5 \leq t < 30.5$	40	$(30.5, 40)$
$31-35$	$30.5 \leq t < 35.5$	48	$(35.5, 48)$
$36-40$	$35.5 \leq t < 40.5$	50	$(40.5, 50)$

NOTE: We also have to plot $(0.5, 0)$ so that the curve will start from the horizontal axis.



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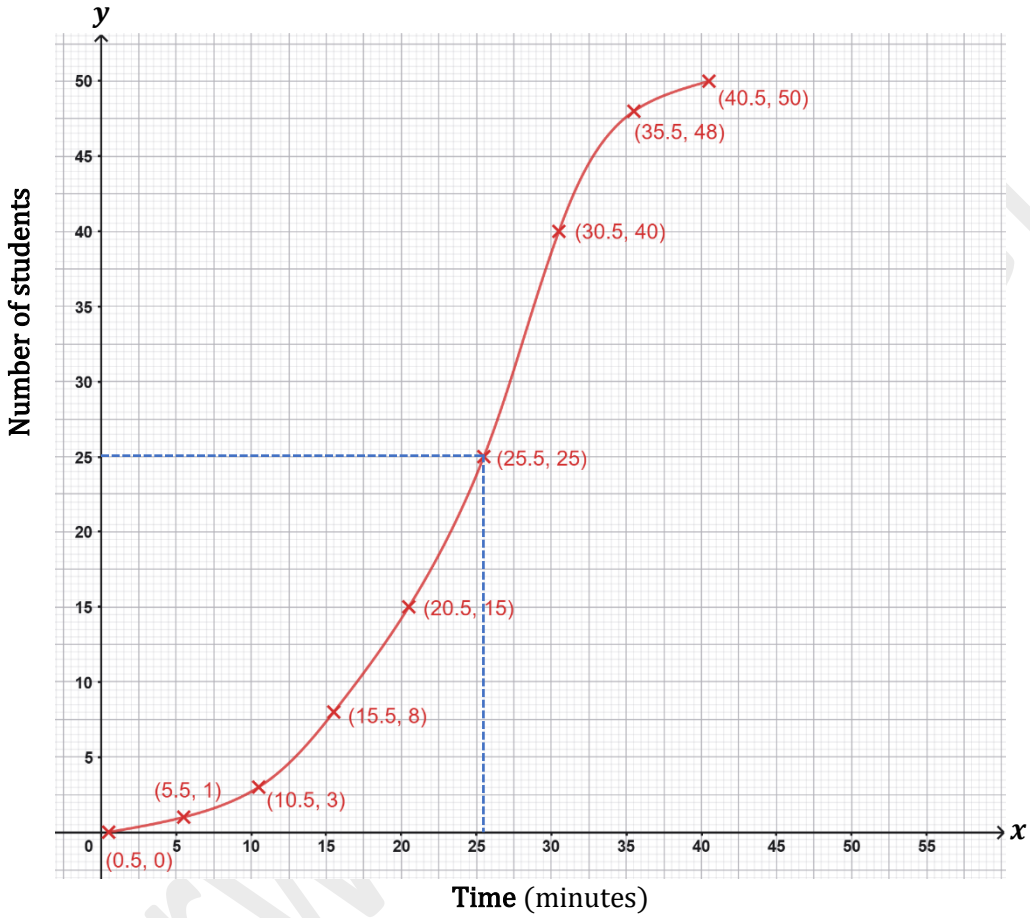
(c) Using the graph, estimate

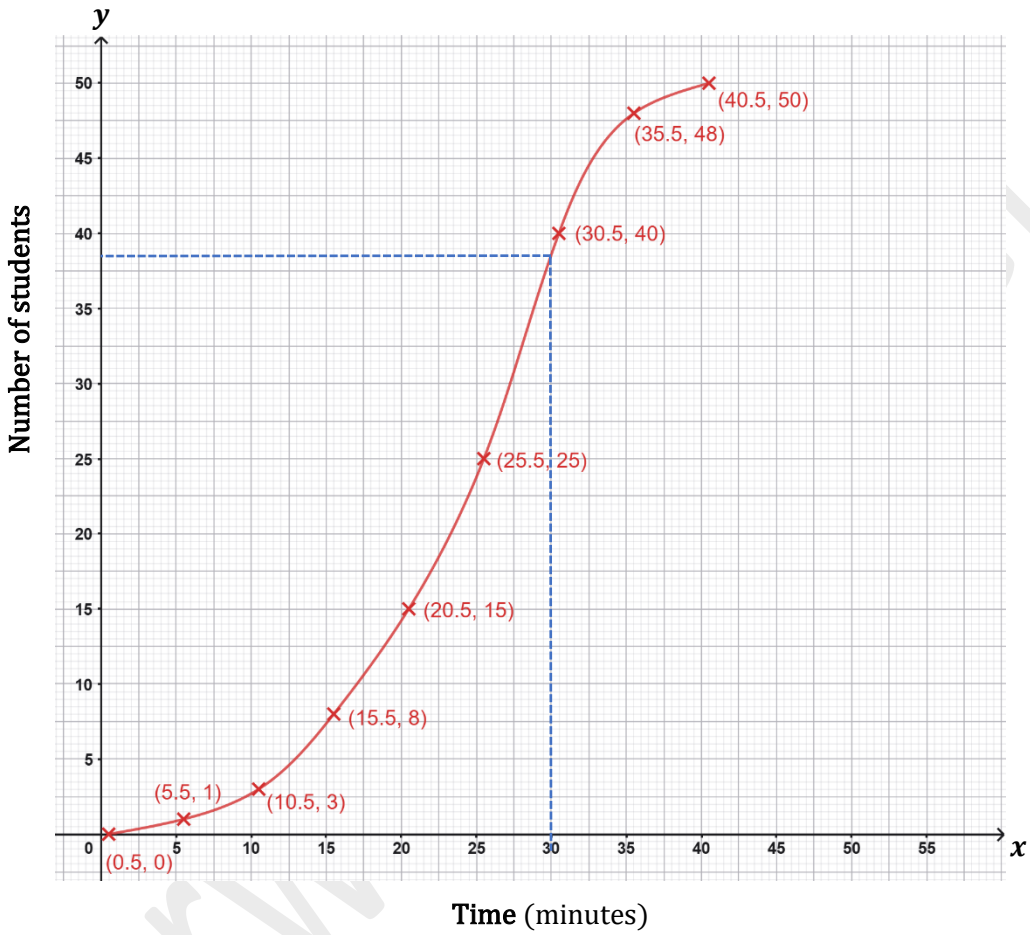
(i) the median time taken to complete the experiment [2]



The median time taken to complete the experiment was 25.5 minutes.

(ii) the probability that a student, chosen at random, took **30 minutes or less** to complete the experiment. [2]

Show on your graph, using broken lines, how these estimates were obtained.



Number of students who took 30 minutes or less = 38.5 students

≈ 39 students

∴ The probability that a student took 30 minutes or less

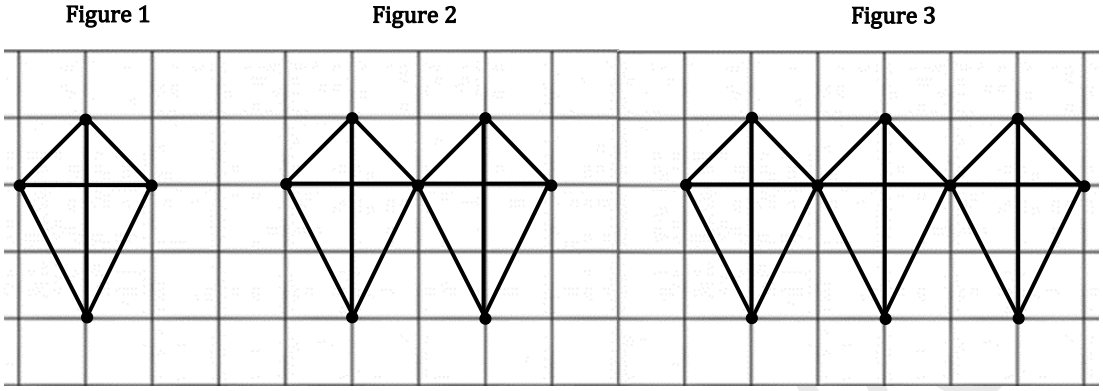
$$= \frac{Number\ of\ students\ who\ took\ 30\ minutes\ or\ less}{Total\ number\ of\ students}$$

$$= \frac{39}{50}$$

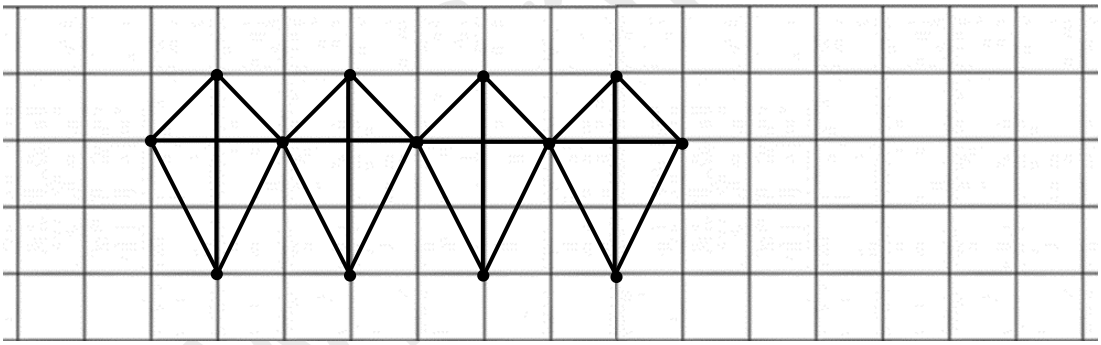
$$= 0.78$$

Total: 11 marks

8. The diagram below shows the first three figures in a sequence of figures.



(a) Draw the fourth figure in the sequence. [2]



(b) The table below shows the number of dots and lines in each figure. Study the pattern in the table and complete the table by inserting the missing values in the rows numbered (i), (ii), (iii) and (iv).

	Figure	Number of Dots	Number of Lines	
	1	4	6	
	2	7	11	
	3	10	16	
(i)	4	13	21	[2]
		Entries omitted for Figures 5-9		
(ii)	10	31	51	[2]
		Entries omitted for some Figures		
(iii)	16	49	81	[2]
		Entries omitted for some Figures		
(iv)	N	$3n + 1$	$5n + 1$	[2]

(iv) For Figure n, the number of dots $= 3n + 1$

For Figure n, the number of lines $= 5n + 1$

Explanation for Number of Dots:

Figure	Number of Dots	Number of Lines
1	4	6
2	7	11
3	10	16

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The number of dots increase by 3.

$\therefore$ In terms of n, we can deduce that $Number\ of\ Dots = 3n + k$

[where k is a constant]

Substituting $n = 1$ and $Number\ of\ Dots = 4$ into:

$Number\ of\ Dots = 3n + k$

$4 = 3(1) + k$

$4 = 3 + k$

$4 - 3 = k$

$1 = k$

$k = 1$

We can also substitute $n = 2$ and Number of Dots $= 7$ OR $n = 3$ and Number of Dots $= 10$ to confirm the value of k.

$\therefore$ Number of Dots $= 3n + 1$

Explanation for Number of Lines:

Figure	Number of Dots	Number of Lines
1	4	6
2	7	11
3	10	16

The number of lines increase by 5.

$\therefore$ In terms of n, we can deduce that $Number\ of\ Lines = 5n + k$

[where k is a constant]

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Substituting $n = 1$ and $Number\ of\ Lines = 6$ into:

$Number\ of\ Lines = 5n + k$

$6 = 5(1) + k$

$6 = 5 + k$

$6 - 5 = k$

$1 = k$

$k = 1$

We can also substitute $n = 2$ and Number of Lines $= 11$ OR $n = 3$ and Number of Lines $= 16$ to confirm the value of k.

$\therefore$ Number of Lines $= 5n + 1$

(i) When $n = 4$,

Number of Dots $= 3n + 1$

$= 3(4) + 1$

$= 12 + 1$

$= 13$

Number of Lines $= 5n + 1$

$= 5(4) + 1$

$= 20 + 1$

$= 21$

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(ii) When $n = 10$,

$$\text{Number of Dots} = 3n + 1$$
$$= 3(10) + 1$$
$$= 30 + 1$$
$$= 31$$

$$\text{Number of Lines} = 5n + 1$$
$$= 5(10) + 1$$
$$= 50 + 1$$
$$= 51$$

(iii) Number of Dots = 49,

$$49 = 3n + 1$$
$$49 - 1 = 3n$$
$$48 = 3n$$
$$\frac{48}{3} = n$$
$$n = \frac{48}{3}$$
$$n = 16$$

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When $n = 16$,

Number of Lines $= 5n + 1$

$= 5(16) + 1$

$= 80 + 1$

$= 81$

Total: 10 marks

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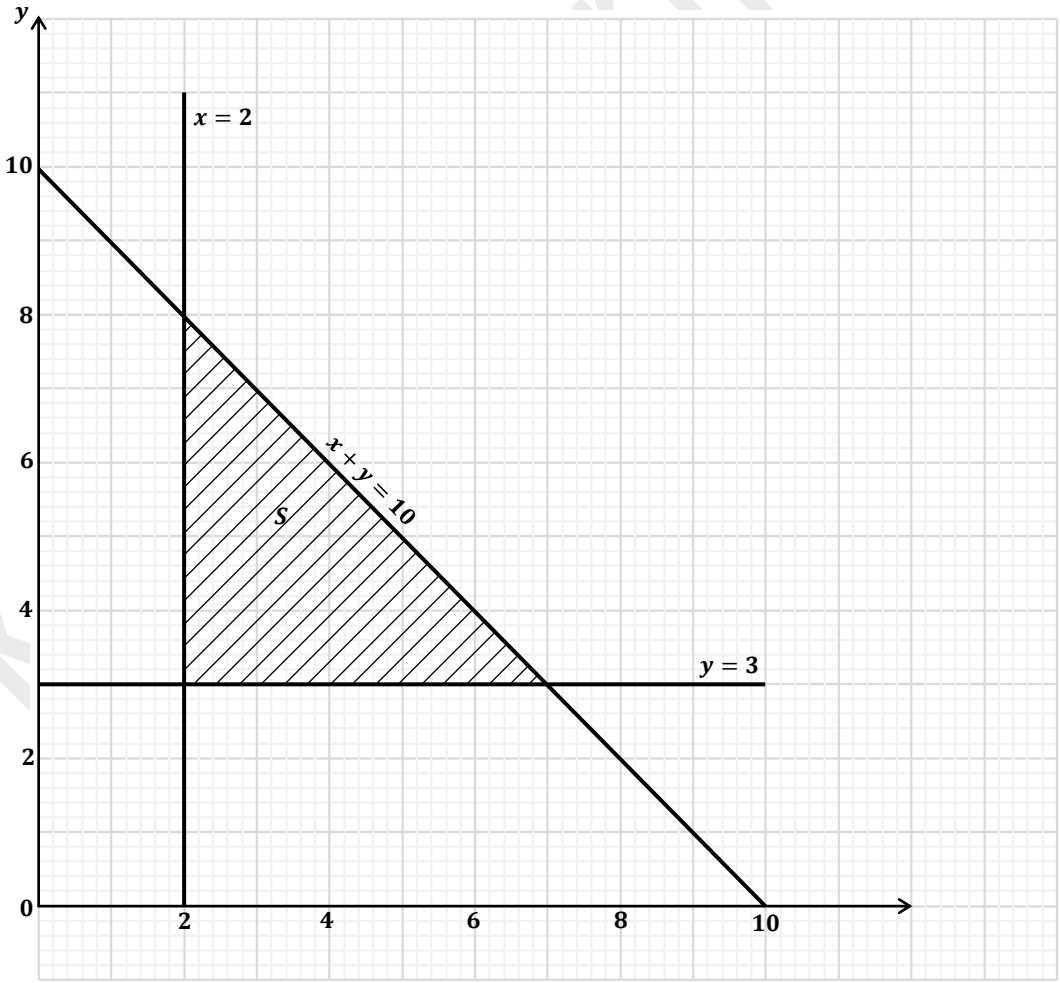
SECTION II

Answer TWO questions in this section.

ALGEBRA AND RELATIONS, FUNCTIONS AND GRAPHS

9. (a) The diagram below shows the graph of three lines and a shaded region, S, defined by three inequalities associated with these lines.

The inequality associated with the line $y = 3$ is $y \geq 3$.



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(i) State the other TWO inequalities which define the shaded region. [2]

The other two inequalities are:

1. $x \geq 2$
2. $x + y \leq 10$

The function $P = 5x + 2y - 3$ satisfies the solution set represented by the closed triangular region.

(ii) Identify the three pairs of (x, y) values for which P has a maximum or a minimum value. [2]

The three pairs of values are $(2, 3)$, $(2, 8)$ and $(7, 3)$.

(iii) Which pair of (x, y) values makes P a **maximum?** [3]

Justify your answer.

$P = 5x + 2y - 3$

Subtituting $(2, 3)$ into P:

$P = 5(2) + 2(3) - 3$

$= 10 + 6 - 3$

$= 13$

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Subtituting $(2, 8)$ into P:

$P = 5(2) + 2(8) - 3$

$= 10 + 16 - 3$

$= 23$

Subtituting $(7, 3)$ into P:

$P = 5(7) + 2(3) - 3$

$= 35 + 6 - 3$

$= 38$

$\therefore$ The point $(7, 3)$ makes P a maximum.

P is a maximum at ……$(7, 3)$…… .

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(b) The function $f(x)$ and $g(x)$ are defined as follows

$f(x) = \frac{3}{2x+1}$ and $g(x) = x^2$

(i) Evaluate EACH of the following:

- $g\left(\frac{-1}{2}\right)$

$$g\left(\frac{-1}{2}\right) = \left(\frac{-1}{2}\right)^2$$

$$= \frac{1}{4}$$

- $fg\left(\frac{-1}{2}\right)$

$$f\left[g\left(\frac{-1}{2}\right)\right] = f\left[\frac{1}{4}\right]$$

$$= \frac{3}{2\left(\frac{1}{4}\right) + 1}$$

$$= 2$$

[4]

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(ii) Write an expression in x for $f^{-1}(x)$. [4]

$$f(x) = \frac{3}{2x+1}$$

Let $y = f(x)$:

$$y = \frac{3}{2x+1}$$

Make x the subject of the formula:

$$y = \frac{3}{2x+1}$$

$$y(2x+1) = 3$$

$$2xy + y = 3$$

$$2xy = 3 - y$$

$$x = \frac{3-y}{2y}$$

Interchange x and y:

$$y = \frac{3-x}{2x}$$

$$\therefore f^{-1}(x) = \frac{3-x}{2x}$$

Total: 15 marks

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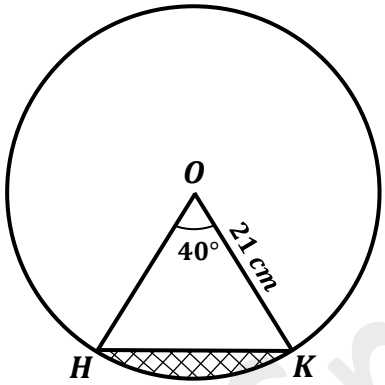
10. (a) The figure below, **not drawn to scale**, shows a circle with centre O.

The radius of the circle is 21 cm and angle $HOK = 40°$.



Use $\pi = \frac{22}{7}$

Determine

(i) the area of the minor sector HOK [2]

$$Area\ of\ the\ minor\ sector\ HOK = \frac{\theta}{360°} \times \pi r^2$$

$$= \frac{40°}{360°} \times \frac{22}{7} \times (21)^2$$

$$= 154\ cm^2$$

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(ii) the area of triangle HOK [3]

$$Area\ of\ triangle\ HOK = \frac{1}{2}\ (OH)(OK)\sin O$$

$$= \frac{1}{2}(21)(21)\sin 40°$$

$$= 141.73\ cm^2 \quad \text{[correct to 2 decimal places]}$$

(iii) the area of the shaded segment [2]

$$Area \text{ of the shaded segment} = \text{Area of minor sector} - \text{Area of triangle}$$

$$= 154\ cm^2 - 141.73\ cm^2$$

$$= 12.27\ cm^2$$

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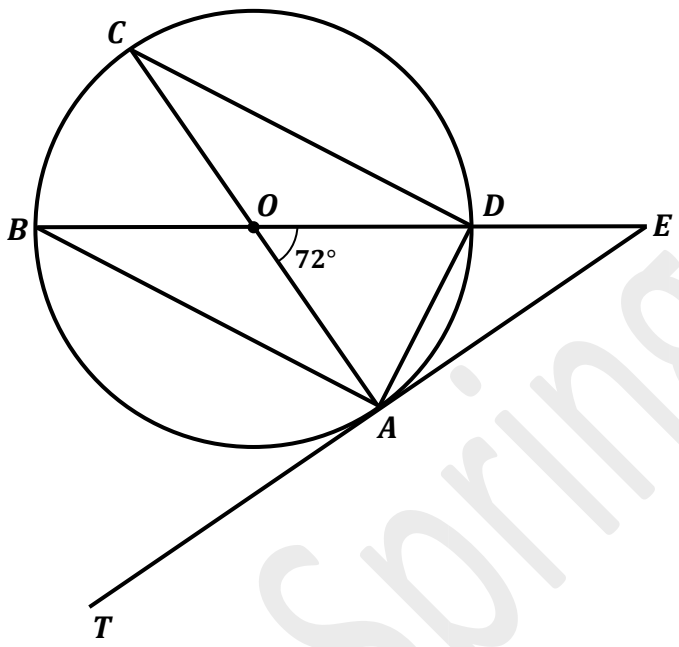
(b) The diagram below, **not drawn to scale**, shows a circle with centre O. TAE is a tangent to the circle at point A and angle $AOD = 72°$.



Calculate, **giving the reason for each step of your answer**, the measure of:

(i) $\angle ADC$ [2]

$A\hat{D}C = 90°$

Since CA is a diameter, the angle standing on a diameter at the circumference is a right angle.

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(ii) $\angle ACD$ [2]

$\angle ACD$ is an angle at the circumference standing on the same arc AD.

The angle at the centre is twice the angle at the circumference standing on the same arc.

$$\therefore \ \angle ACD = \frac{72^\circ}{2}$$

$$= 36^\circ$$

(iii) $\angle CAD$ [2]

The angles in a triangle add up to 180°.

So,

$$\angle CAD = 180^\circ - (\angle ADC + \angle ACD)$$

$$= 180^\circ - (90^\circ + 36^\circ)$$

$$= 54^\circ$$

(iv) $\angle OEA$ [2]

The radius OA meets tangent AE at A.

A radius and a tangent at the point of contact are perpendicular.

$$\therefore \ \angle OAE = 90^\circ$$

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O, D, and E are on the same straight line, so

$\angle AOE = \angle AOD = 72°$

Since angles in a triangle add up to 180°,

$\angle OEA = 180° - (90° + 72°)$

$= 18°$

Total: 15 marks

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11. (a) The points A, B and C have coordinates $A(-2, 8)$, $B(4, 2)$ and $C(0, 9)$.

M is the midpoint of the line segment AB.

(i) Express EACH of the following in the form $\begin{pmatrix} x \\ y \end{pmatrix}$:

- $\overrightarrow{OB} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$

$\overrightarrow{OB} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$

- $\overrightarrow{AB} = \begin{pmatrix} 6 \\ -6 \end{pmatrix}$

$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$

$= -\begin{pmatrix} -2 \\ 8 \end{pmatrix} + \begin{pmatrix} 4 \\ 2 \end{pmatrix}$

$= \begin{pmatrix} 2 \\ -8 \end{pmatrix} + \begin{pmatrix} 4 \\ 2 \end{pmatrix}$

$= \begin{pmatrix} 6 \\ -6 \end{pmatrix}$

- $\overrightarrow{OM} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$

$\overrightarrow{AM} = \frac{1}{2}\overrightarrow{AB}$

$= \frac{1}{2}\begin{pmatrix} 6 \\ -6 \end{pmatrix}$

$= \begin{pmatrix} 3 \\ -3 \end{pmatrix}$

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$$\overrightarrow{OM} = \overrightarrow{OA} + \overrightarrow{AM}$$

$$= \begin{pmatrix} -2 \\ 8 \end{pmatrix} + \begin{pmatrix} 3 \\ -3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

[5]

(ii) Using a vector method, show that $\overrightarrow{AC}$ and $\overrightarrow{OB}$ are parallel. [2]

$$\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC}$$

$$= -\begin{pmatrix} -2 \\ 8 \end{pmatrix} + \begin{pmatrix} 0 \\ 9 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ -8 \end{pmatrix} + \begin{pmatrix} 0 \\ 9 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\overrightarrow{AC} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\overrightarrow{OB} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

$$= 2\begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$= 2\overrightarrow{AC}$$

Since $\overrightarrow{OB} = 2\overrightarrow{AC}$ then $\overrightarrow{OB}$ and $\overrightarrow{AC}$ are parallel.

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(b) The matrix M is defined as $M = \begin{pmatrix} 2p & -3 \\ 4 & 1 \end{pmatrix}$.

Determine the value of p for which the matrix M is singular. [2]

If M is singular then it does not have an inverse and $\det M = 0$

$\det M = (2p \times 1) - (-3 \times 4)$

$2p - (-12) = 0$

$2p + 12 = 0$

$2p = -12$

$p = -\frac{12}{2}$

$p = -6$

(c) A and B are two 2×2 matrice such that $A = \begin{pmatrix} 1 & 2 \\ -4 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & -1 \\ 0 & 3 \end{pmatrix}$.

(i) Calculate $2A + B$. [2]

$2\begin{pmatrix} 1 & 2 \\ -4 & 3 \end{pmatrix} + \begin{pmatrix} 5 & -1 \\ 0 & 3 \end{pmatrix}$

$= \begin{pmatrix} 2 & 4 \\ -8 & 6 \end{pmatrix} + \begin{pmatrix} 5 & -1 \\ 0 & 3 \end{pmatrix}$

$= \begin{pmatrix} 2+5 & 4+(-1) \\ -8+0 & 6+3 \end{pmatrix}$

$= \begin{pmatrix} 7 & 3 \\ -8 & 9 \end{pmatrix}$

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(ii) Determine B^{-1}, the inverse of B. [2]

$$B = \begin{pmatrix} 5 & -1 \\ 0 & 3 \end{pmatrix}$$

$$\det B = (5 \times 3) - (-1 \times 0)$$

$$= 15 - 0$$

$$= 15$$

$$adj\ B = \begin{pmatrix} 3 & 1 \\ 0 & 5 \end{pmatrix}$$

$$B^{-1} = \frac{1}{\det B} \times adj\ B$$

$$= \frac{1}{15}\begin{pmatrix} 3 & 1 \\ 0 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3}{15} & \frac{1}{15} \\ 0 & \frac{5}{15} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{5} & \frac{1}{15} \\ 0 & \frac{1}{3} \end{pmatrix}$$

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(iii) Given that $\begin{pmatrix} 5 & -1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 9 \\ 3 \end{pmatrix}$, calculate the values of x and y. [2]

<u>Recall:</u>

$B = \begin{pmatrix} 5 & -1 \\ 0 & 3 \end{pmatrix}$

<u>Recall:</u>

$B^{-1} \times B = I$

$$B^{-1} \times B \times \begin{pmatrix} x \\ y \end{pmatrix} = B^{-1} \times \begin{pmatrix} 9 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{5} & \frac{1}{15} \\ 0 & \frac{1}{3} \end{pmatrix}\begin{pmatrix} 5 & -1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{5} & \frac{1}{15} \\ 0 & \frac{1}{3} \end{pmatrix} \times \begin{pmatrix} 9 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{5} & \frac{1}{15} \\ 0 & \frac{1}{3} \end{pmatrix} \times \begin{pmatrix} 9 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \left(\frac{1}{5} \times 9\right) + \left(\frac{1}{15} \times 3\right) \\ (0 \times 9) + \left(\frac{1}{3} \times 3\right) \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$\therefore x = 2$ and $y = 1$

Total: 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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